

Bangladesh University of Business & Technology (BUBT)



Lab Report

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Course Code : CSE102

Lab Report No : 02

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Question 1 : Write a C program to read and print elements of array. – using recursion.

```
#include<stdio.h>

int main(){

    int n,x;

    int arr[n];

    printf("Enter size of array: ");

    scanf("%d",&n);

    printf("Enter %d elements of array: ",n);


    for(x=0;x<n;x++){

        scanf("%d",&arr[x]);

    }

    printf("\nElements in array are: ");


    for(x=0;x<n;x++){

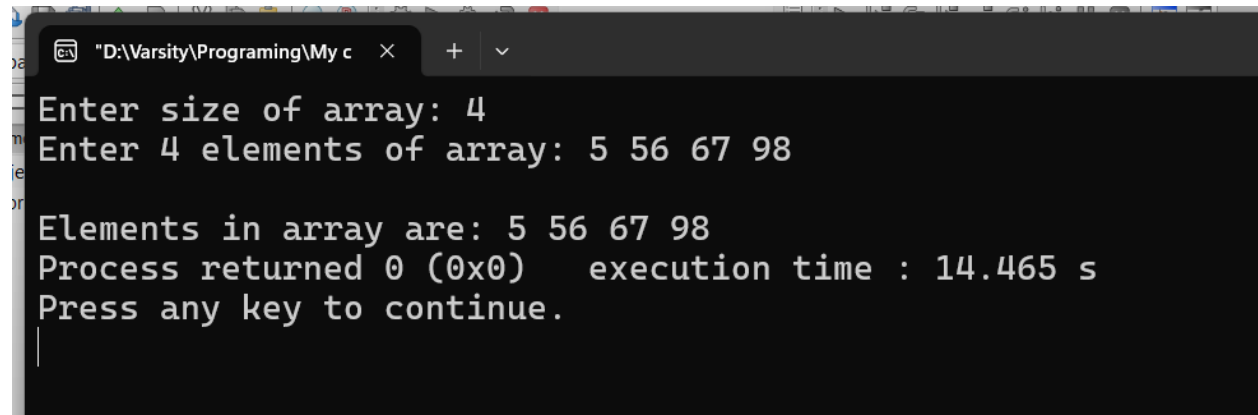
        printf("%d ",arr[x]);

    }

    return 0;

}
```

Output:

A screenshot of a Windows command prompt window titled "D:\Varsity\Programing\My c". The window shows the execution of a C program. The user is prompted to enter the size of the array, which is 4. Then, the user is prompted to enter 4 elements of the array, which are 5, 56, 67, and 98. The program then prints out the elements of the array: 5, 56, 67, and 98. It also displays the process return code as 0 (0x0) and the execution time as 14.465 seconds. Finally, it prompts the user to press any key to continue, which is indicated by a vertical line at the end of the line.

```
"D:\Varsity\Programing\My c" X + v
Enter size of array: 4
Enter 4 elements of array: 5 56 67 98

Elements in array are: 5 56 67 98
Process returned 0 (0x0)   execution time : 14.465 s
Press any key to continue.
|
```

Question 2: Write a C program to print all negative elements in an array.

```
#include<stdio.h>

int main(){
    int n,x;
    int arr[n];

    printf("Enter size of array: ");
    scanf("%d",&n);
    printf("Enter %d elements of array: ",n);

    for(x=0;x<n;x++){
        scanf("%d",&arr[x]);
    }

    printf("\nNegative elements in array are: ");
```

```

for(x=0;x<n;x++){
    if(arr[x]<0){
        printf("%d ",arr[x]);
    }
}

return 0;
}

```

Output:

```

D:\Varsity\Programing\My c
Enter size of array: 4
Enter 4 elements of array: -77 89 -98 455

Negative elements in array are: -77 -98
Process returned 0 (0x0)   execution time : 14.062 s
Press any key to continue.
|

```

Question 3: Write a C program to find sum of all array elements. – using recursion.

```

#include<stdio.h>

int main(){
    int n,x,sum=0;

    int arr[n];

    printf("Enter size of array: ");
    scanf("%d",&n);

```

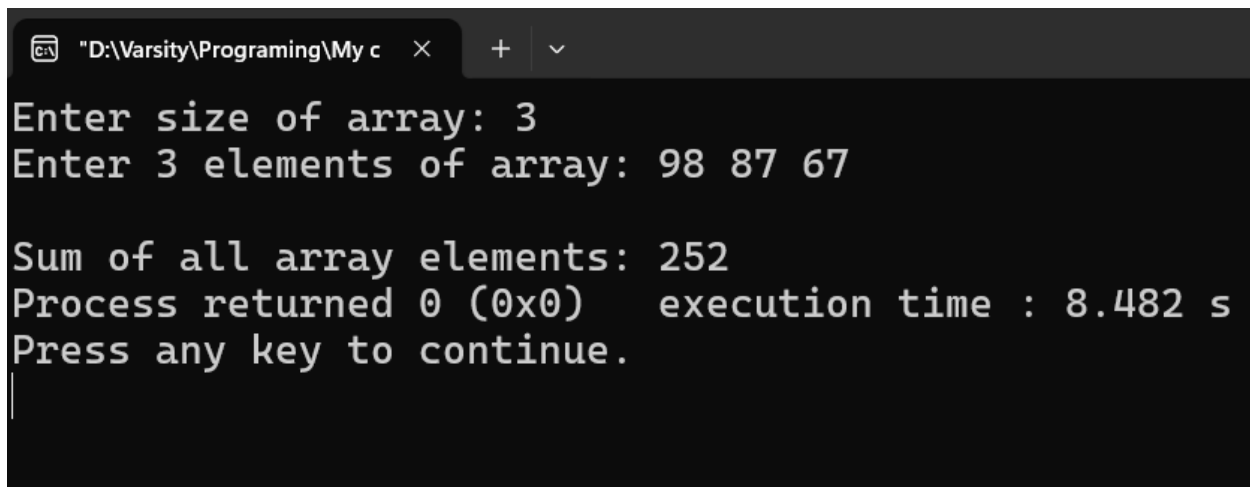
```
printf("Enter %d elements of array: ",n);

for(x=0;x<n;x++){
    scanf("%d",&arr[x]);
    sum=sum+arr[x];
}

printf("\nSum of all array elements: %d",sum);

return 0;
}
```

Output :

A screenshot of a Windows command prompt window titled "D:\Varsity\Programing\My c". The window shows the execution of a C program. The user enters "3" for the array size and "98 87 67" for the array elements. The program outputs the sum of the elements as "252". It also displays "Process returned 0 (0x0) execution time : 8.482 s" and "Press any key to continue." before the cursor moves to a new line.

```
"D:\Varsity\Programing\My c" × + ▾
Enter size of array: 3
Enter 3 elements of array: 98 87 67

Sum of all array elements: 252
Process returned 0 (0x0) execution time : 8.482 s
Press any key to continue.
|
```

Question 4: Write a C program to find maximum and minimum element in an array. – using recursion.

```
#include<stdio.h>

int main(){
```

```
int n,x,max,min;

int arr[n];

printf("Enter size of array: ");

scanf("%d",&n);

printf("Enter %d elements of array: ",n);


for(x=0;x<n;x++){

    scanf("%d",&arr[x]);

}

max=arr[0];

min=arr[0];

for(x=1;x<n;x++){

    if(max<arr[x])max=arr[x];

    if(min>arr[x])min=arr[x];

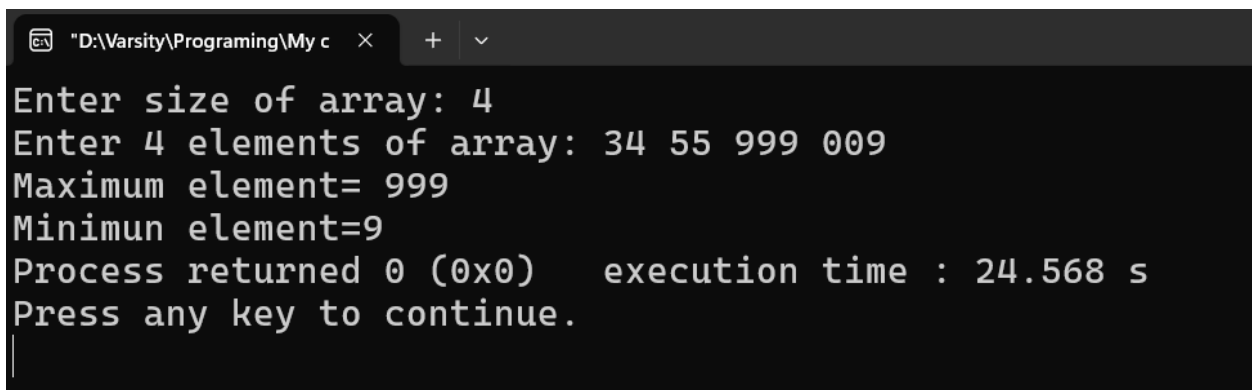
}

printf("Maximum element= %d\nMinimun element=%d",max,min);

return 0;

}
```

Output:



```
"D:\Varsity\Programing\My c"
Enter size of array: 4
Enter 4 elements of array: 34 55 999 009
Maximum element= 999
Minimun element=9
Process returned 0 (0x0)    execution time : 24.568 s
Press any key to continue.
|
```

Question 5:Linear search

```
#include<stdio.h>

int main(){

    int n,x,s,found;

    int arr[n];

    printf("Enter size of array: ");

    scanf("%d",&n);

    printf("Enter %d elements of array: ",n);


    for(x=0;x<n;x++){

        scanf("%d",&arr[x]);

    }

    printf("Enter the element to search: ");

    scanf("%d",&s);

    for(x=0;x<n;x++){

        if(s==arr[x]){

            printf("This element found at position: %d",x);

            found=1;

            break;

        }

    }

    if(!found){

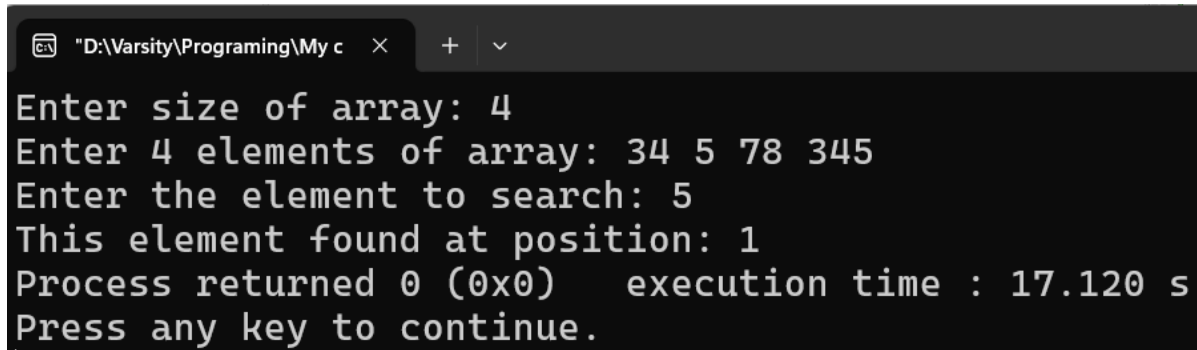
        printf("Not found.");

    }

}
```

```
    return 0;  
}
```

Output:



The screenshot shows a terminal window with a dark background. The title bar at the top indicates the file path "D:\Varsity\Programing\My c" and includes standard window controls. The terminal displays the following text: "Enter size of array: 4", "Enter 4 elements of array: 34 5 78 345", "Enter the element to search: 5", "This element found at position: 1", "Process returned 0 (0x0) execution time : 17.120 s", and "Press any key to continue." followed by a vertical cursor line.

```
Enter size of array: 4  
Enter 4 elements of array: 34 5 78 345  
Enter the element to search: 5  
This element found at position: 1  
Process returned 0 (0x0) execution time : 17.120 s  
Press any key to continue.  
|
```